

Retail Customer Insights & Behavior Analysis

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2. Dataset Summary

- Rows: 3,900
- Columns: 18

Key Features:

- Customer demographics (Age, Gender, Location, Subscription Status)
- Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
- Shopping behavior (Discount Applied, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
- Missing Data: Some missing values observed in the Review Rating column

3. Exploratory Data Analysis using Python

We performed data preparation and analysis using Python:

- **Data Loading:** Imported dataset using pandas
- **Initial Exploration:** Used `df.info()` and `df.describe()` to understand structure and statistics

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	NaN
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	NaN
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	22
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN

Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
3900	3900	3900.000000	3900	3900
2	2	NaN	6	7
No	No	NaN	PayPal	Every 3 Months
2223	2223	NaN	677	584
NaN	NaN	25.351538	NaN	NaN
NaN	NaN	14.447125	NaN	NaN
NaN	NaN	1.000000	NaN	NaN
NaN	NaN	13.000000	NaN	NaN
NaN	NaN	25.000000	NaN	NaN
NaN	NaN	38.000000	NaN	NaN
NaN	NaN	50.000000	NaN	NaN

- Missing Value Handling: Filled missing values in Review Rating using median values
- Data Cleaning: Standardized column names and ensured consistency
- Feature Engineering: Created useful insights from existing columns

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in MySQL to answer key business questions:

We analyzed customer data using SQL to extract key business insights, including:

- Revenue comparison by gender
- Identification of high-spending customers using discounts
- Top-performing products based on customer ratings
- Comparison of purchase behavior across different shipping methods
- Customer segmentation based on purchase history

1. **Revenue by Gender** – Compared total revenue generated by male vs. female customers.

	Gender	revenue
▶	Male	155438
	Female	75191

2. **High-Spending Discount Users** – Identified customers who used discounts but still spent above the average purchase amount.

	customer_id	Purchase Amount (USD)
▶	2	64
	3	73
	4	90
	7	85
	9	97
	12	68
	13	72
	16	81
	20	90
	22	62
	24	88
	29	94
	32	79
	33	67
	35	91
	37	60

3. **Top 5 Products by Rating** – Found products with the highest average review ratings.

	Item Purchased	avg_rating
▶	Gloves	3.86
	Sandals	3.84
	Boots	3.82
	Hat	3.8
	Skirt	3.79

4. **Shipping Type Comparison** – Compared average purchase amounts between Standard and Express shipping.

	Shipping Type	avg_purchase
▶	Express	60.34
	Standard	58.41

5. **Subscribers vs. Non-Subscribers** – Compared average spend and total revenue across subscription status.

	Subscription Status	total_customers	avg_spend	total_revenue
▶	No	2830	59.81	169252
	Yes	1033	59.42	61377

6. **Discount-Dependent Products** – Identified 5 products with the highest percentage of discounted purchases.

	Item Purchased	discount_rate
▶	Hat	49.67
	Sneakers	49.66
	Coat	48.43
	Sweater	47.20
	Pants	46.43

7. **Customer Segmentation** – Classified customers into New, Returning, and Loyal segments based on purchase history.

	customer_segment	total_customers
►	Loyal	3085
	Returning	695
	New	83

8. **Top 3 Products per Category** – Listed the most purchased products within each category.

	item_rank	Category	Item Purchased	total_orders
►	1	Accessories	Jewelry	170
	2	Accessories	Sunglasses	159
	3	Accessories	Belt	159
	1	Clothing	Blouse	169
	2	Clothing	Shirt	168
	3	Clothing	Pants	168
	1	Footwear	Sandals	159
	2	Footwear	Shoes	147
	3	Footwear	Sneakers	145
	1	Outerwear	Jacket	163
	2	Outerwear	Coat	159

9. **Repeat Buyers & Subscriptions** – Checked whether customers with >5 purchases are more likely to subscribe.

	Subscription Status	repeat_buyers
►	Yes	939
	No	2503

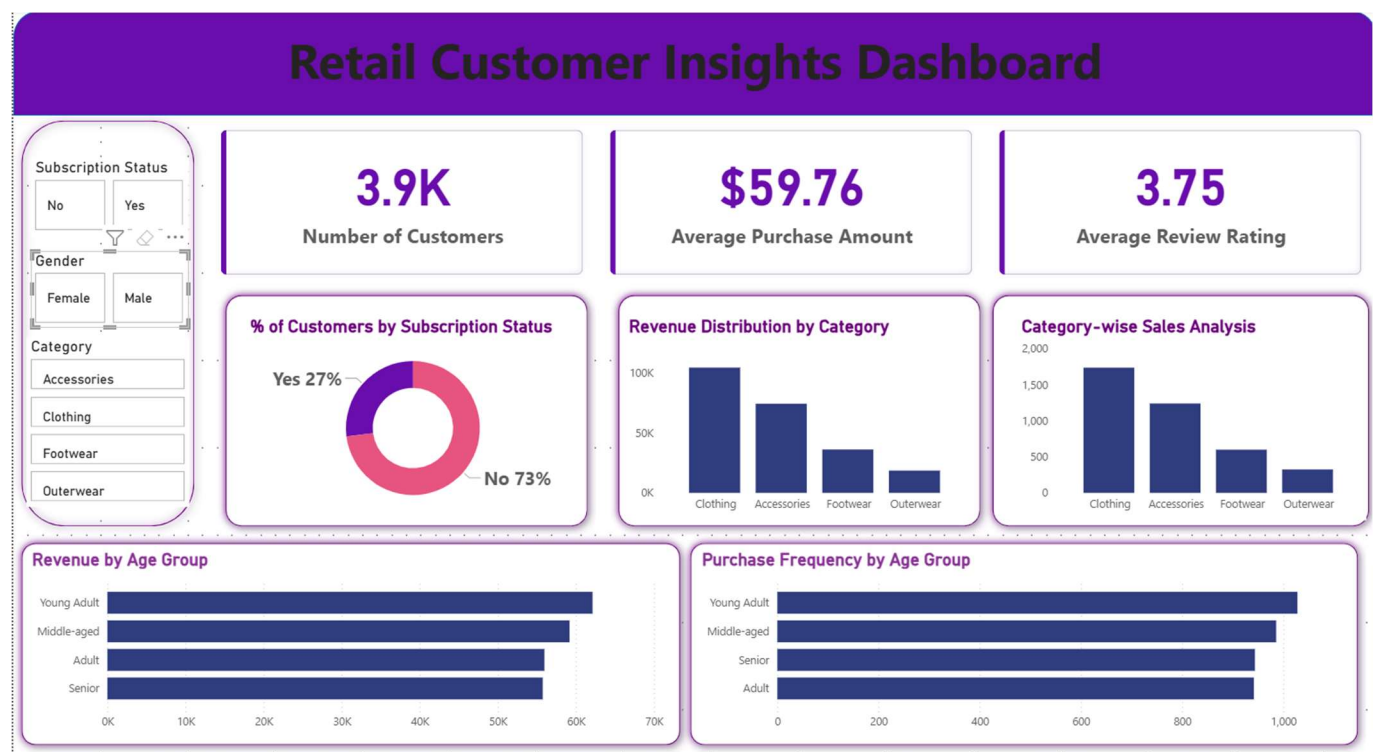
10. **Revenue by Age Group** – Calculated total revenue contribution of each age group.

	age_group	total_revenue
▶	Middle-aged	88465
	Adult	65335
	Senior	42548
	Young Adult	34281

Sample query outputs are shown above to support the analysis.

5. Dashboard in Power BI

Finally, we built an interactive dashboard in **Power BI** to present insights visually.



6. Business Recommendations

- Improve subscription programs to increase customer retention

- Focus marketing on high-value customer segments
- Optimize discount strategies to balance revenue and profit
- Promote top-performing products and categories
- Use seasonal trends for targeted campaigns